

TRACE — Final Six-Presenter Pitch Script & Stage Playbook

Target Duration: 4 Minutes 10 Seconds • **Format:** 6 Presenters × 6 Slides • **Live Prototype Synchronization Included**

Slot & Presenter	Slide & Focus	Time Window	Key On-Screen Action / Demo Cue
Presenter 1 Lead Anchor	Slide 1: Title Page Problem Statement & Vision Hook	0:00 – 0:30 (30 sec)	Show Slide 1; Project confident stance, establish case complexity hook.
Presenter 2 Investigation Context	Slide 2: Problem, Idea & Solution The 5 Gaps & The Quiet Coordinator	0:30 – 1:10 (40 sec)	Show Slide 2 diagram highlighting fragmented records & bridge actor.
Presenter 3 Technical Presenter 1	Slide 3: Technical Approach Pipeline, Graph Engine & grounded QA	1:10 – 2:05 (55 sec)	LIVE DEMO CUE: Switch to Workspace, rotate 3D graph, click AI query.
Presenter 4 Technical Presenter 2	Slide 4: Feasibility & Responsible AI Offline-first, SHA-256 & BSA safeguards	2:05 – 2:55 (50 sec)	Point to Non-Guilt Priority Score, Section 65B notice & audit trail.
Presenter 5 Operational Impact	Slide 5: Impact and Benefits Days to minutes, mule detection, LEAs	2:55 – 3:35 (40 sec)	Show Slide 5 metrics; recite ONLY 2 key impact metrics orally.
Presenter 6 Research & Closer	Slide 6: Research Grounding Louvain, link prediction & closing vision	3:35 – 4:10 (35 sec)	Show Slide 6; deliver final vision quote; open Q&A; smoothly.

Presenter 1 • Slide 1: Title Page

Time: 0:00 – 0:30 (30s)

“Good morning, respected judges.

We are Team TRACE, presenting for Problem Statement SIH26189: AI-Powered Criminal Network Analysis System.

In a serious investigation, one case may contain an FIR, thousands of call records, bank transactions, vehicle information, location trails, and digital evidence. Each file may carry an important clue. But when these clues remain separated, the criminal network remains hidden.

TRACE gives investigators one connected view of the case. It helps them move from isolated records to evidence-linked investigative leads.”

[ACTION / DEMO CUE]: Stand tall, make eye contact with all judges. Hold presentation remote. Set confident, authoritative tone.

HANDOVER CALLOUT: “To understand why this matters, I invite Presenter 2 to explain the investigation gap we are solving.”

Presenter 2 • Slide 2: Problem, Idea, and Solution**Time: 0:30 – 1:10 (40s)**

“Today, investigators spend valuable time manually cross-checking records across multiple files and systems. This creates five major challenges: disconnected data, hidden links, timeline errors, invalid or inconsistent records, and privacy risks during file movement.

A quiet coordinator may never appear in the most call records. However, that person may connect two separate groups, control a financial route, or coordinate activity through intermediaries. Manual analysis can miss such relationships.

TRACE brings FIRs, CDRs, financial transactions, locations, vehicles, documents, and supporting evidence into one investigation workspace.

It extracts relevant entities, connects them through a knowledge graph, validates the available evidence, and gives investigators a traceable path back to the source record.

Our core idea is simple: TRACE converts fragmented criminal data into connected, explainable intelligence.”

[ACTION / DEMO CUE]: Emphasize 'Quiet Coordinator' concept. Gesture to Slide 2's diagram illustrating isolated files vs unified graph.

HANDOVER CALLOUT: “Now, Presenter 3 will show how this intelligence workflow works technically.”

Presenter 3 (Technical Presenter 1) • Slide 3: Technical Approach**Time: 1:10 – 2:05 (55s)**

“TRACE follows an end-to-end technical workflow.

First, investigators upload authorized case records such as FIR text, call-detail records, transaction records, vehicle data, and forensic documents.

Second, our parsing and normalization layer extracts entities such as persons, phone numbers, accounts, locations, vehicles, organizations, transactions, and events. The system then creates relationships between these entities.

Third, the graph engine analyses the case network. It identifies key connectors, community clusters, possible money routes, relationship paths, and high-priority investigative leads.

Finally, the frontend presents this information through an interactive 2D and 3D investigation canvas.

In our TRACE workspace, an investigator can ask grounded questions such as: ‘Who connects two criminal clusters?’ or ‘How is this person connected to a specific account or location?’

The system does not provide an unsupported answer. It directs the investigator towards connected evidence available in the case graph.

Our stack uses React and TypeScript for the frontend, FastAPI for backend services, NetworkX for graph analysis, SQLite WAL for local persistence, Three.js WebGL for 3D visualization, and machine-learning and document-parsing tools for structured extraction.”

[ACTION / DEMO CUE]: LIVE DEMO CUE: Open Case Workspace at <http://localhost:3000> -> Rotate 3D graph to demonstrate bridge nodes -> Click on relationship path -> Point to AI Investigator panel showing source-grounded questions.

HANDOVER CALLOUT: “A criminal-intelligence tool must also be practical, secure, and responsible. Presenter 4 will explain how TRACE meets those requirements.”

Presenter 4 (Technical Presenter 2) • Slide 4: Feasibility, Security, and Responsible AI

Time: 2:05 – 2:55 (50s)

“TRACE is designed for real investigation conditions, including stations with limited infrastructure and unreliable internet.

The prototype has a lightweight architecture that can run on a standard workstation without requiring a GPU cluster. It supports a local-first approach through NetworkX and SQLite WAL, making it suitable for controlled and offline environments.

Real police records are often messy. They may include scanned documents, inconsistent formats, spelling variations, or incomplete fields. TRACE addresses this through OCR cleanup, regex-based normalization, and fuzzy matching before records enter the investigation graph.

We also built responsibility into the design.

Every score is labelled as an investigative priority, never as proof of guilt. The final decision remains with the authorized investigator.

For evidence integrity, TRACE's architecture includes SHA-256 hashing and audit logging for ingested files and actions. The goal is to help the investigator preserve a clear record of where every lead came from.”

[ACTION / DEMO CUE]: On UI, point to the amber banner: 'INVESTIGATIVE DECISION SUPPORT ONLY - Not an automated legal determination'. Show the Audit Trail tab displaying SHA-256 hashes.

HANDOVER CALLOUT: “With a practical and accountable foundation in place, Presenter 5 will explain the impact TRACE can create.”

Presenter 5 • Slide 5: Impact and Benefits**Time: 2:55 – 3:35 (40s)**

“TRACE can help investigation teams spend less time searching through records and more time validating meaningful leads.

For a complex case involving multiple files, manual linking can take days. TRACE creates an initial connected view of the case within minutes, allowing investigators to focus their time on verification, field action, and prosecution preparation.

Its graph analytics can reveal low-profile bridge actors, mule-account patterns, coordinated criminal cells, and hidden financial relationships.

Its timeline and geo-radar functions help investigators connect actions across time and location. When a call, a transaction, a vehicle movement, and an intercepted conversation align, the case becomes easier to understand and verify.

Potential users include investigating officers, cybercrime units, state crime-record bureaus, special task forces, and central law-enforcement agencies.”

[ACTION / DEMO CUE]: STAGE RULE FOR SLIDE 5: Mention only TWO impact points orally (e.g. reduction from days to minutes, and unmasking mule accounts). Do NOT read raw statistics from the slide.

HANDOVER CALLOUT: “Finally, Presenter 6 will explain the research foundation behind TRACE and close our presentation.”

Presenter 6 • Slide 6: Research Foundation and Closing**Time: 3:35 – 4:10 (35s)**

“TRACE is grounded in established research, rather than treating AI as a black box.

Our approach draws on criminal-network link prediction, money-laundering network analysis, community-detection algorithms such as Louvain and Leiden, digital-evidence integrity, and research on automation bias in public-sector decision-making.

This research directly shaped our design choices: graph-based reasoning, source-linked evidence, human review, auditability, and investigative-priority labels instead of guilt claims.

TRACE gives investigators a clearer way to understand complex cases. It connects scattered records, reveals meaningful relationships, and supports accountable action.

Our vision is simple: every investigator should be able to trace a lead back to the evidence that supports it.

Thank you. We are Team TRACE, and we are ready for your questions.”

[ACTION / DEMO CUE]: STAGE RULE FOR SLIDE 6: Do NOT read paper titles or authors. Explain WHY research makes TRACE credible and trustworthy. Deliver closing quote with conviction.

HANDOVER CALLOUT: “[CLOSE OF PITCH • TRANSITION TO JUDGES Q&A;]”

TRACE // STAGE EXECUTION RULES & DEFENSE PROTOCOLS

Strict guidelines for all 6 presenters during the presentation and live demonstration rounds:

Rule #	Guideline & Stage Constraint	Exact Mandate / Rebuttal Script
Rule 1	Technical Duo Alignment	Presenter 3 (Workflow/Demo) and Presenter 4 (Feasibility/Security) operate as a synchronized pair. While P3 explains the graph pipeline, P4 coordinates on the laptop/mouse.
Rule 2	No Verbatim Reading	Do not read slide text word-for-word. The slides provide visual validation; your spoken voice provides narrative impact, clarity, and authority.
Rule 3	Slide 5 Oral Limit	Mention ONLY two impact points orally (e.g. 'days-to-minutes synthesis' and 'mule account unmasking'). Do not recite every percentage and statistic shown on the slide.
Rule 4	Slide 6 Research Explanation	Do not read paper titles or author citations. Explain WHY the research matters: it proves TRACE avoids black-box hallucination and prevents automation bias.
Rule 5	Prototype & Synthetic Data Notice	When presenting or showing the live workspace, state clearly: <i>"This demonstration uses synthetic case data in our prototype."</i>
Rule 6	Non-Guilt Determination Mandate	NEVER state that TRACE 'proves guilt', 'catches criminals automatically', or 'replaces human investigators'. Always state: <i>'Generates ranked investigative leads for authorized human verification.'</i>
Rule 7	No False Deployment Claims	NEVER claim TRACE already has official government deployment. State: <i>'Built as an operational prototype aligned with Indian law enforcement standards (BSA, BNSS, Sec 65B).'</i>
Rule 8	Hosted Site vs Backend Roadmap	If judges ask about the live hosted site (Netlify/Vercel): <i>"The current hosted build demonstrates the client-side workflow and synthetic cases; secure backend integration is part of our production roadmap."</i>

JUDGES Q&A; DEFENSE CHEAT-SHEET

Anticipated Judge Question	Designated Responder & Bulletproof Defense Answer
Q1: How do you prevent hallucination or accusing innocent citizens?	Presenter 4 / P6: <i>"TRACE uses graph-grounded extraction rather than generative generation. Every node and link has a strict cryptographic citation to a specific line in an FIR, CDR, or bank log. If an edge cannot cite a source file, it cannot appear in the graph. Scores are strictly labelled as 'Investigative Priority' under human review."</i>
Q2: What if a rural police station has no internet or GPUs?	Presenter 4: <i>"TRACE is engineered local-first. It runs on a standard Core i5 / 8GB RAM police desktop using NetworkX and SQLite WAL with zero cloud or GPU dependencies. Ingested files never leave the precinct premises."</i>
Q3: Is digital evidence acceptable in Indian courts?	Presenter 4 / P1: <i>"Yes. TRACE incorporates automated Section 65B Indian Evidence Act / Section 63 BSA statutory certifications and generates SHA-256 tamper-evident cryptographic chain of custody logs for every ingested file and transcript."</i>